HealthConnect Rwanda

Healthcare Operations and Performance: Business Insights Report

AnalystLab Africa Experience Lab, Week 6, Data Analytics track | Yawa Jacqueline GBEDEVI | October 2026

1. Summary

The data describes a network of 20 facilities that looks unusually balanced. Patients split almost evenly between men and women and between the five provinces. Visits are spread over every month of 2026 with no real season. Five diagnoses each make up about a fifth of cases, and each insurer and each payment method holds roughly a quarter of its total. Real hospital networks are rarely this even, so the first honest finding is that the dataset gives few signals of strain, and the signals it does give are on the money side.

Five points matter most for management.

- **Collections.** 9.31 billion was billed across 60,000 bills and 8.39 billion collected (90.1%). That leaves 920 million outstanding (9.9%), spread over 11,887 bills that still carry a balance.
- **Facilities.** HF011, HF006 and HF013 bill the most among appointment-linked bills. HF006 and HF012 collect best (92.2%). HF014 is fifth on billing but nineteenth on collection (88.5%).
- **Emergency load.** One visit in three is an emergency (1,650 of 5,000), which is high enough to deserve a review of emergency capacity.
- **Staffing.** Every facility has 61 or 64 staff although visits vary by 28% between facilities, so the busiest ones carry the most visits per staff member.
- **Data gaps.** Only 5,000 of the 60,000 appointments referenced by billing exist in the file. Every analysis by facility, department or month therefore covers about 8% of the bills.

2. Data and approach

The 27 source tables were loaded into PostgreSQL, left untouched in a raw schema, and cleaned into a separate schema with 27 documented quality checks. Forty SQL queries produced the figures below, and a six-page Power BI dashboard presents them. Amounts are shown as recorded in the dataset (assumed to be Rwandan francs).

The quality checks changed what can be concluded, so they come first.

- **Missing appointments.** The appointment table stops at APT005000, while billing, diagnoses and feedback run to APT060000. This was confirmed against the source file.
- **Missing tables.** There are no admissions, laboratory, prescription or referral records, no facility names and no registration dates. Admission rate, length of stay, referrals, lab and prescription analysis are therefore out of reach.
- **Inconsistent links.** For 95% of appointments the department recorded does not belong to the facility recorded. Departments and facilities are analysed separately for that reason.
- **Billing status.** Every bill marked Pending has a payment recorded, and some bills marked Paid still have a balance. Outstanding amounts are computed as billed minus collected, whatever the label says.
- **Other issues.** Phone numbers were corrupted at the source (10 distinct values for 20,000 patients), and the monthly revenue summary table (1.13 billion) does not reconcile with billing (9.31 billion) or payments (8.39 billion). Neither is used for conclusions.

3. Findings

3.1 Patients

There are 20,000 registered patients, 10,090 men and 9,910 women. The average age is 45 and ages run from 0 to 90. Each province holds between 19.6% and 20.5% of patients (Western 4,091, Eastern 4,067, Southern 3,967, Kigali City 3,952, Northern 3,923). Every patient carries an insurance ID, so coverage reads as 100%. The IDs fall into 15 codes of 1,270 to 1,379 patients each, and none of them looks like a "no insurance" value. With no reference table to name the codes, they are best read as insurance schemes or plans, not as proof that every patient is covered in practice.

3.2 Activity

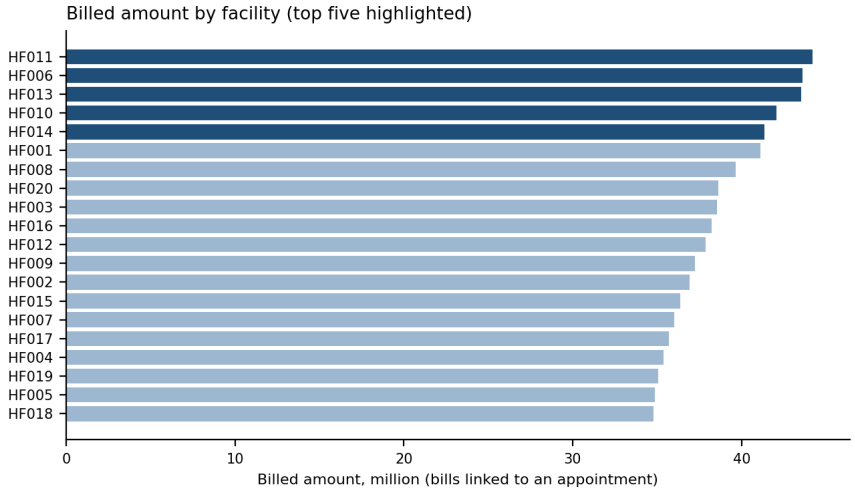
The 5,000 available appointments are spread flat across the year, from 400 a month (May) to 440 (March), about 417 on average. The busiest facilities are HF013 (278 visits), HF014 (276), HF010 (275), HF001 and HF011 (273 each). HF019 is the quietest with 217, so the busiest facility sees about 28% more visits than the least busy one. Visit types are close to thirds, with 1,694 follow-ups, 1,656 consultations and 1,650 emergencies. The 350 doctors average about 14 appointments each over the year, a very low figure that probably reflects the missing appointment records.

3.3 Clinical picture

Across 60,000 diagnoses, five conditions share the load almost equally. Gastritis is the most frequent at 20.3%, followed by diabetes (20.1%), hypertension (20.1%), malaria (19.8%) and upper respiratory infection (19.6%). Diabetes and hypertension together account for 40% of cases, which matters for long-term care planning. Severity is also split in thirds, with 33.3% of cases recorded as severe.

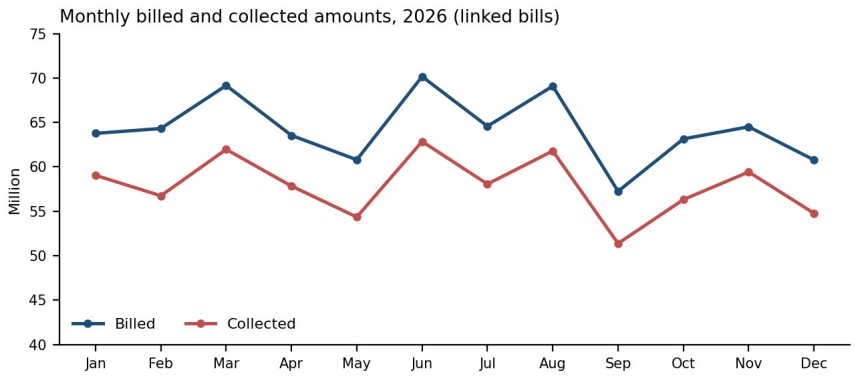
3.4 Revenue and collection

The average bill is 155,157, with a median of 155,247. Amounts range from 10,002 to 299,998 and are spread evenly across that range, so no handful of large bills drives revenue. Among the 5,000 bills linked to appointments, the five leading facilities carry 27.9% of the billed amount, which is only a little above an even share of 25%.



Facility	Bills	Billed (million)	Collection rate	Overall rank (revenue and collection)
HF011	273	44.2	90.1%	3 (tied)
HF006	269	43.6	92.2%	1
HF013	278	43.5	90.7%	2
HF010	275	42.1	90.2%	5
HF014	276	41.4	88.5%	12
HF007	224	36.0	88.2%	19
HF019	217	35.0	89.5%	16 (tied)
HF005	223	34.9	88.9%	20
HF018	240	34.8	89.8%	16 (tied)

Revenue follows visit volume more than price. The highest and lowest facilities differ by 27% in billed amount, while average bills sit in a narrow band of 145,000 to 163,000. By department, Cardiology generates the most (11.2% of the billed amount), ahead of Radiology (10.6%) and Laboratory (10.2%). Pharmacy is lowest at 9.4%, so the whole range is under two points.



Monthly billing moves between 57.2 million (September) and 70.2 million (June). The second half of the year is 3.2% below the first half, and the fourth quarter 4.5% below the first. On 5,000 bills this is a gentle drift, not evidence of a decline, but it is worth watching once the missing appointments are loaded.

3.5 Insurance and payments

The 42,036 claims, worth 4.88 billion, are shared almost equally among four insurers. Radiant leads with 10,644 claims and 1.237 billion claimed, followed by RSSB (10,511), Private (10,508) and MMI (10,373). Claimed amounts differ by less than 2% from one insurer to the next. Approval rates sit between 49.3% (Private) and 50.9% (RSSB). The remaining claims are pending, and none is marked as rejected, which suggests rejections are not being recorded. Payments are also evenly split, with card (25.2%), mobile money (25.0%), insurance (24.9%) and cash (24.8%).

About 10% of the billed amount is outstanding whatever the bill status, which confirms the status label cannot be used to chase payments. Paid bills carry 311 million of balance, partially paid ones 306 million and pending ones 303 million.

3.6 Staffing

Staffing is almost identical from one facility to the next. Each has 17 or 18 doctors, 35 nurses, 4 administrative staff and a few pharmacists, laboratory scientists and radiographers, which makes 61 or 64 people per facility and 1,250 across the network (350 of them doctors). Visits, in contrast, vary by 28% between facilities, so the load per staff member is uneven. The five busiest facilities handle 4.3 to 4.6 visits per staff member in the available data, against about 3.5 at the quietest ones.

3.7 Patient experience

Ratings are spread evenly over the 1 to 5 scale, with about 4,150 to 4,270 reviews per score. Only 1,756 of the 21,064 reviews can be tied to a facility. Average ratings by facility run from 3.19 (HF011) down to 2.76 (HF007 and HF020), with HF001 (3.12) and HF012 (3.08) just behind the leader. With 72 to 102 reviews per facility, gaps of this size could easily arise by chance, so the ranking is a prompt for questions and not a verdict.

Facility	Visits	Staff	Visits per staff	Average rating	Reviews linked
HF013	278	61	4.56	2.77	96
HF014	276	61	4.52	2.77	102
HF010	275	64	4.30	2.85	97
HF001	273	64	4.27	3.12	93
HF011	273	61	4.48	3.19	91
HF019	217	61	3.56	2.97	76
HF004	222	64	3.47	2.99	83
HF005	223	64	3.48	2.82	74

One pattern deserves attention anyway. HF013 and HF014, the two busiest facilities, both sit at 2.77, and HF014 also collects less than average (88.5%). HF011 is just as busy and has the best rating, so crowding alone does not explain the scores.

4. The ten business questions

Question	Answer from the data
Which hospitals serve the most patients?	HF013, HF014, HF010, HF001 and HF011 (273 to 278 visits each). HF019 is lowest with 217. Based on the 5,000 available appointments.
Which departments carry the greatest workload?	Differences are small. Cardiology has the most bills (544), then Radiology (525) and Obstetrics & Gynecology (511). ICU is lowest (467).
Which diseases are becoming more common?	This cannot be answered reliably. The five diagnoses each hold about 20% of cases, and only 5,000 diagnoses carry a date, which gives roughly 80 records per diagnosis and month, too few to call a trend.
Are medical staff distributed efficiently?	Not in proportion to demand. Every facility has 61 or 64 staff (17 or 18 doctors, 35 nurses) while visits range from 217 to 278. Visits per staff member run from about 3.5 at the quietest facilities to 4.6 at HF013.
Which insurers generate the largest revenue?	Radiant, with 10,644 claims and 1.237 billion claimed. On claimed amount, the four insurers are within 2% of each other.
Which facilities have the highest patient satisfaction?	HF011 (3.19), HF001 (3.12) and HF012 (3.08) rate highest, HF007 and HF020 (2.76) lowest. The gap is small, with 72 to 102 reviews per facility, and could be chance.
Which hospitals require expansion?	The data cannot prove it, since every bed is recorded as available and no occupancy exists. HF013, HF014 and HF010 come first for review, with the most visits per staff member. HF013 and HF014 are also among the lowest rated (2.77).
How can operational costs be reduced?	No cost data exists. The measurable lever is on the revenue side, with 920 million (9.9%) billed but not yet collected.
Which services contribute most to revenue?	Bills carry no service field, so departments stand in. Cardiology (11.2%), Radiology (10.6%) and Laboratory (10.2%) lead.
Which regions need more investment?	No region stands out. Patients are spread between 19.6% and 20.5% per province, and facilities have no location in the data.

5. Recommendations

- **Tighten collections first.** The outstanding 920 million is the largest quantified opportunity. Start with the 11,887 bills that carry a balance, and look first at the facilities collecting below 89% (HF007, HF014, HF017, HF005). Billing status should then be driven by actual payments so that follow-up lists can be trusted.
- **Match staffing to demand.** Staff numbers are nearly the same everywhere while visits are not. Moving a few nurses or doctors toward HF013, HF014, HF010, HF001 and HF011 (273 to 278 visits against 217 at the quietest facility) is a cheaper first step than building capacity. Bed occupancy should be checked before any expansion decision.
- **Investigate HF014 and HF013.** They are the two busiest facilities, both rated 2.77. HF014 also collects below average. A short review of waiting times, billing practice and staffing there would show whether the three signals share a cause.
- **Look at the emergency pathway.** A third of visits are emergencies. If part of them are routine problems arriving through the emergency door, a triage or follow-up clinic would free emergency capacity. Recording the reason for each emergency visit would settle the question.
- **Plan for chronic care.** Diabetes and hypertension make up 40% of diagnoses. Structured follow-up programmes for these two would reach a large share of patients.
- **Set processing targets with insurers.** About half of all claims are still pending with every insurer. Start with Radiant, which holds the largest volume, and confirm that rejected claims are being recorded at all.
- **Fix the data capture.** Load the 55,000 missing appointments, add facility names and locations, and add admissions, laboratory, prescription and referral tables. That unlocks the questions this report could not answer, namely expansion needs, disease trends, regional gaps and satisfaction by hospital.

6. Limits of this analysis

Every facility, department and monthly figure rests on the 5,000 bills that can be tied to an appointment, about 8% of billing. Totals for billed, collected and outstanding amounts, claims and payment methods use all 60,000 bills. Satisfaction averages rest on 72 to 102 reviews per facility. Differences between facilities are modest and could shift once the full appointment history is available. Rankings are best read as a starting list for investigation, not as a verdict on any hospital.